Derivation: Qubit Encoding and the Jordan-Wigner Transform

Seminar notes

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1 Setup: The Jordan-Wigner Transform

The goal is to map the fermionic Kitaev chain to a system of qubits. Fermionic operators obey the anticommutation relations:

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = 0. \tag{1}$$

Qubit operators on different sites commute: $[\sigma_i^\alpha, \sigma_j^\beta] = 0$ for $i \neq j$. To simulate fermions with qubits, we must encode the fermionic anticommutation into the qubit algebra. This is achieved using the Jordan-Wigner transform, which attaches a "string" of Pauli Z operators to each fermionic lowering operator to act as a parity counter for the preceding sites:

$$c_j = \left(\prod_{k=0}^{j-1} Z_k\right) \sigma_j^-, \qquad c_j^\dagger = \left(\prod_{k=0}^{j-1} Z_k\right) \sigma_j^+. \tag{2}$$

Here, $\sigma^\pm = \frac{1}{2}(X \pm iY)$. The operators X, Y, Z are the standard Pauli matrices acting on the two-level qubit system at site j:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3}$$

These matrices satisfy the standard Pauli algebra $X^2 = Y^2 = Z^2 = I$, anticommute with each other ($\{X, Y\} = 0$, etc.), and form the basis for observing the state of the qubit. The Z-string flips sign when an earlier site is occupied (because $Z|1\rangle = -|1\rangle$), restoring the correct anticommutation relations between operators at different sites.

2 Derivation of the Qubit Hamiltonian

We start with the Kitaev Hamiltonian in open boundary conditions:

$$H_K = -\mu \sum_{j=0}^{L-1} c_j^\dagger c_j - t \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1} + \text{h.c.}) + \Delta \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1}^\dagger + \text{h.c.}). \tag{4}$$

We apply the Jordan-Wigner mapping term by term.

2.1 Chemical Potential Term

The number operator at site j maps to the single-qubit Z basis:

$$c_j^\dagger c_j = \sigma_j^+ \sigma_j^- = \left(\frac{X_j + iY_j}{2}\right)\left(\frac{X_j - iY_j}{2}\right) = \frac{I + Z_j}{2}. \tag{5}$$

Summing this yields the transverse field term:

$$-\mu \sum_j c_j^\dagger c_j = -\frac{\mu}{2}\sum_j (I + Z_j) \cong -\frac{\mu}{2}\sum_j Z_j \quad \text{(dropping constant offsets).} \tag{6}$$

Note: With the operator convention used here ($c_j^\dagger = (\prod_{k<j} Z_k)\,\sigma_j^+$, $\sigma^\pm = (X \pm iY)/2$) one has $c_j^\dagger c_j = \sigma_j^+\sigma_j^- = \frac{I+Z_j}{2}$, so the *occupied* single-site state is $|0\rangle$ ($Z = +1$). This is the convention the code (`src/jordan_wigner.py`) uses, and it reproduces the $-\frac{\mu}{2}\sum_j Z_j$ transverse field written in the slides directly, with no sign flip.

2.2 Hopping Term

For the hopping term $c_j^\dagger c_{j+1}$, the Z-strings partially cancel:

$$\begin{aligned} c_j^\dagger c_{j+1} &= \left(\prod_{k=0}^{j-1} Z_k\right)\sigma_j^+ \cdot \left(\prod_{m=0}^{j} Z_m\right)\sigma_{j+1}^- \\ &= \sigma_j^+ Z_j \sigma_{j+1}^- = \sigma_j^+(2\sigma_j^+\sigma_j^- - I)\sigma_{j+1}^- \\ &= -\sigma_j^+\sigma_{j+1}^-. \end{aligned}$$

Adding its Hermitian conjugate:

$$c_j^\dagger c_{j+1} + \text{h.c.} = -(\sigma_j^+\sigma_{j+1}^- + \sigma_j^-\sigma_{j+1}^+) = -\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2}. \tag{7}$$

Multiplying by $-t$, we get the XX and YY coupling.

2.3 Pairing Term

Similarly, for the pairing term $c_j^\dagger c_{j+1}^\dagger$:

$$c_j^\dagger c_{j+1}^\dagger = \sigma_j^+ Z_j \sigma_{j+1}^+ = -\sigma_j^+\sigma_{j+1}^+.$$

Adding its Hermitian conjugate:

$$c_j^\dagger c_{j+1}^\dagger + \text{h.c.} = -(\sigma_j^+\sigma_{j+1}^+ + \sigma_j^-\sigma_{j+1}^-) = -\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2}. \tag{8}$$

Multiplying by Δ, this generates an asymmetric XX and YY coupling.

2.4 Full Qubit Hamiltonian

Combining all terms, we obtain the Kitaev Hamiltonian in the qubit basis:

$$\begin{aligned} H &= -\frac{\mu}{2}\sum_j Z_j + t\sum_j \left(\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2}\right) - \Delta \sum_j \left(\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2}\right) \\ &= -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}. \end{aligned} \tag{9}$$

This is the transverse-field XY model.

3 The Sweet Spot $t = \Delta$

At the "sweet spot" ($t = \Delta$), the $X_j X_{j+1}$ terms perfectly cancel. The Hamiltonian simplifies to a transverse-field Ising-like model but with YY interactions:

$$H = -\frac{\mu}{2} \sum_j Z_j + t \sum_j Y_j Y_{j+1}. \tag{10}$$

If $\mu = 0$ (deep inside the topological phase), the Hamiltonian is purely $H = t \sum_j Y_j Y_{j+1}$. The boundary operators X_0 and X_{L-1} do not appear in this Hamiltonian, meaning they form zero-energy modes that are exactly decoupled from the bulk. These correspond to the localized Majorana edge modes.

4 Fermion Parity and Topological Degeneracy

4.1 Fermion Parity Operator

The total fermion parity operator is defined as the parity of the number of fermions:

$$P_f = (-1)^{\sum_j c_j^\dagger c_j}. \tag{11}$$

Using the number operator identity $c_j^\dagger c_j = (I + Z_j)/2$, each local factor is $(-1)^{n_j} = -Z_j$. Thus, the physical fermion-parity operator is

$$P_f = (-1)^L \prod_{j=0}^{L-1} Z_j. \tag{12}$$

With the convention used in this project, the code operator

$$\hat{P}_Z = \prod_{j=0}^{L-1} Z_j \tag{13}$$

equals the physical fermion parity $P_f = (-1)^N$ for even L, and differs from it by a minus sign for odd L. All parity-sector plots and VQE claims in the slides use even L, so the $+1$ eigenspace of $\hat{P}_Z$ is the fermion-even sector there.

4.2 Conservation and Parity Gap

Proof of Parity Conservation: The Kitaev Hamiltonian contains three types of terms: the chemical potential ($c_j^\dagger c_j$), the hopping ($c_j^\dagger c_{j+1}$ + h.c.), and the pairing ($\Delta c_j^\dagger c_{j+1}^\dagger$ + h.c.). Crucially, every term is bilinear in the fermionic creation and annihilation operators. The pairing term creates or destroys fermions *in pairs*, while the hopping and chemical potential terms preserve the fermion number. Because fermions are only ever created or destroyed in pairs, the overall parity of the fermion number is strictly conserved.

Mathematically, this means the Hamiltonian commutes with the total parity operator: $[H, P_f] = 0$. It also commutes with the code operator $\hat{P}_Z = \prod_j Z_j$, which differs from P_f only by the fixed sign $(-1)^L$. Since they commute, they share a common set of eigenstates. The 2^L-dimensional many-body Hilbert space block-diagonalizes into two decoupled sectors. For the even chain lengths used in the project, the $\hat{P}_Z = +1$ sector is fermion-even and the $\hat{P}_Z = -1$ sector is fermion-odd; for odd L these labels are reversed.

The Parity Gap: We define the **parity gap** as the energy difference between the absolute ground state of the even sector ($E_0^{(+)}$) and the absolute ground state of the odd sector ($E_0^{(-)}$):

$$\Delta_P = |E_0^{(+)} - E_0^{(-)}|. \tag{14}$$

The behavior of this gap fundamentally distinguishes the trivial and topological phases in the many-body spectrum:

- **Trivial Phase ($|\mu| > 2t$):** The system behaves like a conventional insulator. The lowest energy excitation requires breaking a Cooper pair or overcoming the bulk bandgap. Thus, adding a single fermion to flip the parity costs a macroscopic amount of energy. The parity gap Δ_P is of the order of the bulk gap, $\Delta_{\text{bulk}} \sim |\mu| - 2t$, and remains large even as $L \to \infty$.
- **Topological Phase ($|\mu| < 2t$):** The system hosts localized Majorana zero modes at its boundaries, γ_L and γ_R. These two Majoranas can be combined into a single, highly non-local fermionic mode $f = \frac{1}{2}(\gamma_L + i\gamma_R)$. Because γ_L and γ_R commute with the Hamiltonian in the thermodynamic limit, the occupation of this non-local fermion $f^\dagger f \in \{0, 1\}$ costs strictly zero energy. However, changing the occupation of f flips the total parity of the system. Thus, in the topological phase, the even and odd ground states become perfectly degenerate as $L \to \infty$.

Finite-Size Splitting: In a finite chain of length L, the Majorana wavefunctions γ_L and γ_R are not perfectly localized at the edges; their tails decay exponentially into the bulk with a coherence length $\xi = 1/\ln(2t/|\mu|)$. The overlap of these tails causes the two Majoranas to hybridize, lifting the exact degeneracy. The energy cost to occupy the non-local fermion f becomes non-zero, shifting the odd-parity ground state relative to the even-parity ground state.

This splitting is the finite-size parity gap, which decays exponentially with the system size L:

$$\Delta_P(L) = |E_0^{(+)} - E_0^{(-)}| \approx 2t \left(1 - \left(\frac{|\mu|}{2t}\right)^2\right) \left(\frac{|\mu|}{2t}\right)^L. \tag{15}$$

This exponential vanishing of the parity gap is a direct, observable signature of topological Majorana edge modes in the many-body qubit spectrum.

4.3 Many-Body Qubit Spectrum vs Finite-Size BdG Spectrum (Wavefunction Perspective)

To deeply understand the relationship between the spectra, it is crucial to distinguish the state spaces they operate in: the single-particle BdG wavefunction $|\Psi\rangle$ versus the many-body qubit wavefunction $|\Phi\rangle$.

The BdG Spectrum ($|\Psi\rangle$): The **finite-size BdG spectrum** describes single-particle quasiparticle excitations. A BdG state $|\Psi\rangle$ is a $2L$-dimensional vector $|\Psi\rangle = (u_1, v_1, \ldots, u_L, v_L)^T$ containing the electron and hole amplitudes of a *single* Bogoliubov excitation on top of the implicit BCS ground state. Because of particle-hole symmetry, these excitation energies always come in $\pm E$ pairs. The lowest positive energy $E_0(L)$ represents the exact energy cost to add one single Bogoliubov quasiparticle to the system. In the topological phase, the wavefunctions $|\Psi_L\rangle$ and $|\Psi_R\rangle$ for this lowest excitation correspond to the left and right Majorana modes.

It is important to note that for a finite chain, $|\Psi_L\rangle$ and $|\Psi_R\rangle$ are **not** the true eigenstates of the BdG Hamiltonian. They are localized basis states (ansätze) representing the purely left-localized and right-localized Majorana modes. Because their wavefunctions overlap in the middle of a finite chain, they couple to each other. The *true* eigenstates of the finite-size BdG Hamiltonian are their symmetric and antisymmetric superpositions:

$$|\Psi_+\rangle = \frac{|\Psi_L\rangle + |\Psi_R\rangle}{\sqrt{2}} \quad \text{and} \quad |\Psi_-\rangle = \frac{|\Psi_L\rangle - |\Psi_R\rangle}{\sqrt{2}}. \tag{16}$$

These true eigenstates are spread across both ends of the chain and possess energies $+E_0(L)$ and $-E_0(L)$. It is only in the thermodynamic limit ($L \to \infty$), where the overlap strictly vanishes, that $|\Psi_L\rangle$ and $|\Psi_R\rangle$ become true, decoupled, zero-energy eigenstates.

The Many-Body Qubit Spectrum ($|\Phi\rangle$): The **many-body qubit spectrum** represents the total energies of the full system. A state $|\Phi\rangle$ is a 2^L-dimensional vector in the tensor product space of all L qubits, e.g., $|\Phi\rangle = \sum c_{i_1 \dots i_L} |i_1 \dots i_L\rangle$. This wavefunction describes the collective state of the entire chain.

Because the Hamiltonian commutes with the parity operator ($[H, P] = 0$), every many-body eigenstate $|\Phi\rangle$ has a definite parity. The spectrum splits into:

- **Even sector (blue curves):** States satisfying $P|\Phi_{\text{even}}\rangle = +|\Phi_{\text{even}}\rangle$. The absolute lowest energy state here is the overall system ground state, $|\Phi_0^{(+)}\rangle$, with energy $E_0^{(+)}$.
- **Odd sector (red curves):** States satisfying $P|\Phi_{\text{odd}}\rangle = -|\Phi_{\text{odd}}\rangle$. The lowest energy state here is $|\Phi_0^{(-)}\rangle$, with energy $E_0^{(-)}$.

Ground State Parity: The absolute many-body ground state has **fermion-even parity**. In the $\hat{P}_Z$ labeling used by the code this is the $+1$ sector for even L and the -1 sector for odd L. The system naturally prefers to pair up all its fermions to minimize energy (due to the superconducting pairing term Δ). The lowest fermion-odd state is achieved by breaking a pair or occupying the lowest available hybridized Majorana mode. Because occupying this mode costs a strictly non-zero energy $E_0(L) > 0$ in a finite system, the fermion-odd state usually sits slightly higher in energy at finite L; the two parity sectors become degenerate in the topological phase as $L \to \infty$.

How $|\Psi\rangle$ and $|\Phi\rangle$ Are Related: The link between the $2L$-dimensional single-particle BdG space and the 2^L-dimensional many-body space lies in **second quantization** and the definition of the quasiparticle vacuum.

The BdG eigenvectors $|\Psi_n\rangle = (u_{n,1}, \dots, u_{n,L}, v_{n,1}, \dots, v_{n,L})^T$ do not directly describe physical particles; instead, they define the coefficients of the Bogoliubov quasiparticle creation operators:

$$\gamma_n^\dagger = \sum_{j=1}^{L} \left(u_{n,j} c_j^\dagger + v_{n,j} c_j \right). \tag{17}$$

The absolute many-body ground state of the system, $|\Phi_0^{(+)}\rangle$ (the BCS vacuum), is defined as the state that is annihilated by all Bogoliubov destruction operators: $\gamma_n |\Phi_0^{(+)}\rangle = 0$ for all n.

Excited many-body states $|\Phi\rangle$ are then constructed by acting on this vacuum with the creation operators defined by the BdG wavefunctions. For example, the lowest odd-parity many-body state is created by populating the lowest energy quasiparticle (the hybridized Majorana mode):

$$|\Phi_0^{(-)}\rangle = \gamma_0^\dagger |\Phi_0^{(+)}\rangle. \tag{18}$$

Thus, the BdG wavefunctions $|\Psi_n\rangle$ provide the exact "recipe" (the u and v amplitudes) to build the operators that assemble the full 2^L-dimensional many-body states $|\Phi\rangle$ out of the even-parity ground state vacuum.

Why Even and Odd Parity Curves Do Not Overlap (Finite-Size Effects): To transition from the even-parity ground state $|\Phi_0^{(+)}\rangle$ to the odd-parity ground state $|\Phi_0^{(-)}\rangle$, one must inject exactly one fermion into the system. In the BdG picture, this corresponds to populating the lowest available single-particle excitation $|\Psi_0\rangle$, which costs an energy $E_0(L)$.

Therefore, the energy of the many-body odd state is the energy of the many-body even state plus the single-particle excitation cost:

$$E_0^{(-)} = E_0^{(+)} + E_0(L). \tag{19}$$

In an infinite chain ($L \to \infty$) inside the topological phase, the left and right Majorana BdG wavefunctions $|\Psi_L\rangle$ and $|\Psi_R\rangle$ are infinitely far apart. They do not overlap, resulting in a strictly zero-energy excitation $E_0(\infty) = 0$. In this limit, adding a fermion costs nothing, making the many-body ground states perfectly degenerate: $E_0^{(-)} = E_0^{(+)}$.

However, in a **finite-size** chain, the Majorana wavefunctions $|\Psi_L\rangle$ and $|\Psi_R\rangle$ decay exponentially into the bulk but have a finite overlap. This wavefunction overlap causes the modes to hybridize, pushing the BdG excitation energy away from zero by the splitting amount $E_0(L) \approx 2t(|\mu|/2t)^L$.

Because $E_0(L)$ is strictly non-zero for any finite L, the odd-parity many-body state must lie slightly higher in energy than the even-parity state:

$$\Delta_P(L) = E_0^{(-)} - E_0^{(+)} = E_0(L) > 0. \tag{20}$$

This non-zero cost to populate the hybridized Majorana mode is exactly why the blue (even) and red (odd) curves in the many-body qubit spectrum do not perfectly overlap. The gap between them is precisely the finite-size BdG splitting. As one tunes μ towards the critical point $|\mu| = 2t$, the Majorana wavefunctions delocalize, their overlap becomes macroscopic, the splitting $E_0(L)$ grows, and the parity gap visibly opens.